

MATA37H3 — Week 15

Final Exam Review: Comprehensive

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Roadmap: The Full MATA37H3 Topic List

This review week cycles through every topic from Weeks 1-14: Riemann sums and the definite integral, the Fundamental Theorem of Calculus, integration techniques (substitution, by parts, trig integrals, partial fractions, Weierstrass substitution), improper integrals, numerical integration, sequences, series and the divergence test, convergence tests (Integral, Comparison, LCT, Alternating Series, Ratio), absolute vs. conditional convergence, power series, Taylor polynomials, and Taylor/Maclaurin series with the Binomial Series. No new theorems are introduced this week — this is consolidation and exam practice.

Key Fact: True/False Is a High-Stakes Section

On real MATA37H3-style final exams, a True/False section is often the *first* section and carries significant weight. These questions test precise recall of hypotheses (e.g., “the Ratio Test is inconclusive when the limit equals 1” is true; “the Ratio Test proves divergence when the limit equals 1” is false). A single missing hypothesis (boundedness, continuity, positivity of terms) turns a true statement false. Read every word before answering.

Key Fact: The Convergence-Test Decision Tree

Given a series $\sum a_n$, ask in order: (1) Does $a_n \rightarrow 0$? If not, **Divergence Test** applies immediately. (2) Are all terms positive (or eventually positive)? If the terms resemble $\frac{1}{n^p}$ or fit an integral, try the **Integral Test** or **p-Series** benchmark. If they resemble a known series by comparison, try **Comparison** or **Limit Comparison**. (3) Do the terms alternate in sign? Try the **Alternating Series Test**, and separately ask whether $\sum |a_n|$ converges (**absolute** convergence) or only $\sum a_n$ converges (**conditional**). (4) Does the series involve factorials, n -th powers, or $n!$ -type growth? Try the **Ratio Test**.

Key Fact: The Taylor Series Toolkit

To find a Taylor/Maclaurin series: (a) compute derivatives directly from the coefficient formula $\frac{f^{(k)}(a)}{k!}$; (b) substitute into one of the six core Maclaurin series (e^x , $\sin x$, $\cos x$, $\frac{1}{1-x}$, $\ln(1+x)$, the Binomial Series); or (c) manipulate a known series termwise (differentiate, integrate, multiply by x^k , add/subtract series). To bound the error of a truncated series, use the **Alternating Series Remainder** ($|R_n| \leq b_{n+1}$) when the series alternates and its terms decrease, or the **Lagrange Remainder** ($|R_n(x)| \leq \frac{M}{(n+1)!} |x-a|^{n+1}$) otherwise.

Worked Examples

EXAMPLE 1

Riemann Sums / FTC

Evaluate $\int_1^4 \left(3x^2 - \frac{2}{x}\right) dx$ using the Fundamental Theorem of Calculus.

Solution. An antiderivative is $F(x) = x^3 - 2 \ln x$. Then

$$\int_1^4 \left(3x^2 - \frac{2}{x}\right) dx = F(4) - F(1) = (64 - 2 \ln 4) - (1 - 2 \ln 1) = 63 - 2 \ln 4.$$

Since $\ln 1 = 0$, the final answer is $63 - 2 \ln 4 = 63 - 4 \ln 2$. ■

EXAMPLE 2

Integration by Parts

Evaluate $\int x \ln x \, dx$.

Solution. Let $u = \ln x$, $dv = x \, dx$, so $du = \frac{1}{x} dx$ and $v = \frac{x^2}{2}$. Then

$$\int x \ln x \, dx = \frac{x^2}{2} \ln x - \int \frac{x^2}{2} \cdot \frac{1}{x} dx = \frac{x^2}{2} \ln x - \int \frac{x}{2} dx = \frac{x^2}{2} \ln x - \frac{x^2}{4} + C. \blacksquare$$

EXAMPLE 3

Partial Fractions

Evaluate $\int \frac{5x - 1}{(x - 1)(x + 2)} dx$.

Solution. Write $\frac{5x - 1}{(x - 1)(x + 2)} = \frac{A}{x - 1} + \frac{B}{x + 2}$. Clearing denominators: $5x - 1 = A(x + 2) + B(x - 1)$. At $x = 1$: $4 = 3A \Rightarrow A = \frac{4}{3}$. At $x = -2$: $-11 = -3B \Rightarrow B = \frac{11}{3}$. Thus

$$\int \frac{5x - 1}{(x - 1)(x + 2)} dx = \frac{4}{3} \ln |x - 1| + \frac{11}{3} \ln |x + 2| + C. \blacksquare$$

EXAMPLE 4

Improper Integral

Determine whether $\int_1^\infty \frac{1}{x^{3/2}} dx$ converges, and if so, find its value.

Solution. This is a p -integral with $p = \frac{3}{2} > 1$, so it converges. Compute directly:

$$\int_1^\infty x^{-3/2} dx = \lim_{b \rightarrow \infty} \left[-2x^{-1/2} \right]_1^b = \lim_{b \rightarrow \infty} \left(-\frac{2}{\sqrt{b}} + 2 \right) = 2. \blacksquare$$

EXAMPLE 5*Numerical Integration*

Use Simpson's Rule with $n = 4$ to approximate $\int_0^2 x^3 dx$, and compare to the exact value.

Solution. Here $\Delta x = \frac{2-0}{4} = \frac{1}{2}$, nodes $x_0, \dots, x_4 = 0, 0.5, 1, 1.5, 2$, with $f(x) = x^3$: $f(x_0) = 0$, $f(x_1) = 0.125$, $f(x_2) = 1$, $f(x_3) = 3.375$, $f(x_4) = 8$.

$$S_4 = \frac{\Delta x}{3} [f(x_0) + 4f(x_1) + 2f(x_2) + 4f(x_3) + f(x_4)] = \frac{0.5}{3} [0 + 0.5 + 2 + 13.5 + 8] = \frac{0.5}{3} (24) = 4.$$

The exact value is $\int_0^2 x^3 dx = \left[\frac{x^4}{4} \right]_0^2 = 4$. Simpson's Rule is exact here because it integrates cubics exactly. ■

EXAMPLE 6*Sequence Convergence*

Determine whether $a_n = \frac{3n^2 + 1}{n^2 - 2n}$ converges, and find its limit if so.

Solution. Divide numerator and denominator by n^2 :

$$a_n = \frac{3 + \frac{1}{n^2}}{1 - \frac{2}{n}} \rightarrow \frac{3 + 0}{1 - 0} = 3 \quad \text{as } n \rightarrow \infty.$$

The sequence converges to 3. ■

EXAMPLE 7*Convergence Test Selection*

Determine whether $\sum_{n=1}^{\infty} \frac{n+5}{n^3-2}$ converges.

Solution. For large n , $a_n = \frac{n+5}{n^3-2}$ behaves like $\frac{n}{n^3} = \frac{1}{n^2}$. Apply the Limit Comparison Test with $b_n = \frac{1}{n^2}$:

$$\lim_{n \rightarrow \infty} \frac{a_n}{b_n} = \lim_{n \rightarrow \infty} \frac{n^2(n+5)}{n^3-2} = \lim_{n \rightarrow \infty} \frac{n^3+5n^2}{n^3-2} = 1.$$

Since the limit is a finite positive number and $\sum \frac{1}{n^2}$ is a convergent p -series ($p = 2 > 1$), the given series also converges. ■

EXAMPLE 8*Absolute vs. Conditional Convergence*

Determine whether $\sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{\sqrt{n}}$ converges absolutely, converges conditionally, or diverges.

Solution. The series alternates, with $b_n = \frac{1}{\sqrt{n}}$ decreasing to 0, so by the Alternating Series Test the series converges. Checking absolute convergence: $\sum \left| \frac{(-1)^{n-1}}{\sqrt{n}} \right| = \sum \frac{1}{\sqrt{n}}$ is a p -series with $p = \frac{1}{2} \leq 1$, which diverges. Therefore the series converges **conditionally**. ■

EXAMPLE 9

Power Series Interval of Convergence

Find the radius and interval of convergence of $\sum_{n=1}^{\infty} \frac{(x-3)^n}{n \cdot 4^n}$.

Solution. By the Ratio Test:

$$\lim_{n \rightarrow \infty} \left| \frac{(x-3)^{n+1}}{(n+1)4^{n+1}} \cdot \frac{n \cdot 4^n}{(x-3)^n} \right| = \frac{|x-3|}{4} \lim_{n \rightarrow \infty} \frac{n}{n+1} = \frac{|x-3|}{4}.$$

Convergence requires $\frac{|x-3|}{4} < 1$, i.e. $|x-3| < 4$, so $R = 4$. Checking endpoints: at $x = 7$, the series is $\sum \frac{1}{n}$, which diverges; at $x = -1$, the series is $\sum \frac{(-1)^n}{n}$, which converges (Alternating Series Test). The interval of convergence is $[-1, 7)$. ■

EXAMPLE 10

Taylor Series with Error Bound

Estimate $\cos(0.5)$ using the Maclaurin polynomial of degree 4, and bound the error using the Lagrange Remainder.

Solution. The degree-4 Maclaurin polynomial for $\cos x$ is $p_4(x) = 1 - \frac{x^2}{2} + \frac{x^4}{24}$. At $x = 0.5$:

$$p_4(0.5) = 1 - \frac{0.25}{2} + \frac{0.0625}{24} = 1 - 0.125 + 0.0026 = 0.8776.$$

For the Lagrange bound, $f^{(5)}(t) = -\sin t$, so $|f^{(5)}(t)| \leq 1$ for all t . Thus

$$|R_4(0.5)| \leq \frac{1}{5!}(0.5)^5 = \frac{0.03125}{120} \approx 0.00026.$$

So $\cos(0.5) \approx 0.8776$ with error under 0.0003. ■

Practice Problems

Read the hint and the caution before you start each problem — not after.

1. If $\lim_{n \rightarrow \infty} a_n = 0$, then $\sum_{n=1}^{\infty} a_n$ converges.

HINT Recall the exact logical status of the Divergence Test: $a_n \rightarrow 0$ is required for convergence, but think about whether the converse direction also holds.

CAUTION — before you solve Have a concrete counterexample ready (the harmonic series) rather than just recalling the rule name — exams often ask you to justify, not just answer.

Your working:

2. If the Ratio Test gives $\lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right| = 1$, then $\sum a_n$ diverges.

HINT Recall the three possible outcomes of the Ratio Test and which values of L correspond to each.

CAUTION — before you solve ‘Inconclusive’ is a real, common outcome on exams — don’t round $L = 1$ up to $L > 1$ or down to $L < 1$ in your memory of the rule.

Your working:

3. Every absolutely convergent series is also conditionally convergent.

HINT Write out the precise definitions of ‘absolutely convergent’ and ‘conditionally convergent’ side by side and check whether one set of series is a subset of the

other.

CAUTION — before you solve The two terms describe mutually exclusive categories for a convergent series — a series is one or the other, never both.

Your working:

4. If f is continuous on $[a, b]$, then $\int_a^b f(x) dx$ exists.

HINT Recall the theorem connecting continuity on a closed, bounded interval to Riemann integrability.

CAUTION — before you solve This is about EXISTENCE of the integral (as a number), not about whether you can find an elementary closed-form antiderivative — those are different questions.

Your working:

5. The interval of convergence of a power series is always a closed interval $[a - R, a + R]$.

HINT Recall that the radius of convergence theorem only pins down the open interval $(a - R, a + R)$ — think about what must be checked separately at the two boundary points.

CAUTION — before you solve Endpoint behavior can go three different ways (both diverge, both converge, one of each) — don't assume symmetry between the two endpoints.

Your working:

Part B — Mixed Computation.

6. Evaluate $\int \frac{x}{\sqrt{4-x^2}} dx$ using an appropriate substitution.

HINT Let $u = 4 - x^2$ so that $du = -2x dx$, matching the $x dx$ in the numerator up to a constant.

CAUTION — before you solve Track the sign from $du = -2x dx$ carefully — it flips the sign of the final antiderivative if dropped.

Your working:

7. Determine whether $\int_0^\infty e^{-2x} dx$ converges, and if so, find its value.

HINT Write the improper integral as $\lim_{b \rightarrow \infty} \int_0^b e^{-2x} dx$ and find an antiderivative of e^{-2x} using the chain rule in reverse.

CAUTION — before you solve The antiderivative of e^{-2x} carries a factor of $-\frac{1}{2}$ from the chain rule — a bare $-e^{-2x}$ is off by that factor.

Your working:

8. Use the Trapezoidal Rule with $n = 4$ to approximate $\int_0^4 \sqrt{x} \, dx$.

HINT Set up the Trapezoidal Rule formula $T_n = \frac{\Delta x}{2} [f(x_0) + 2f(x_1) + 2f(x_2) + 2f(x_3) + f(x_4)]$ with $\Delta x = 1$.

CAUTION — before you solve Only the two endpoint function values get weight 1 — every interior node gets weight 2; mixing this up is the most common Trapezoidal Rule error.

Your working:

9. Determine whether $\sum_{n=2}^{\infty} \frac{1}{n(\ln n)^2}$ converges, using the Integral Test.

HINT Substitute $u = \ln x$ (so $du = \frac{dx}{x}$) to convert the integral into a p -integral in u .

CAUTION — before you solve Don't forget to change the lower bound from $x = 2$ to $u = \ln 2$ when you substitute — the bounds transform along with the variable.

Your working:

10. Determine whether $\sum_{n=1}^{\infty} \frac{(-1)^n n}{n^2 + 1}$ converges absolutely, converges conditionally, or diverges.

HINT First check whether $|a_n| \rightarrow 0$; if it does, that alone doesn't decide absolute convergence — use Limit Comparison Test on $|a_n|$ against a simpler benchmark series like $\frac{1}{n}$.

CAUTION — before you solve Reaching ' $|a_n| \rightarrow 0$ ' is not the same as ' $\sum |a_n|$ converges' — don't stop the absolute-convergence check at the term-limit step; you still need a convergence test on the series of absolute values.

Your working:

11. Find the Taylor series for $f(x) = \frac{1}{x}$ centered at $a = 2$, and state its radius of convergence.

HINT Rewrite $\frac{1}{x}$ as $\frac{1}{2+(x-2)}$, then factor out a 2 from the denominator to expose the geometric series form $\frac{1}{1-r}$ with $r = -\frac{x-2}{2}$.

CAUTION — before you solve Don't forget the overall $\frac{1}{2}$ factor left outside the geometric series after factoring — it multiplies every term of the final series.

Your working:

12. Use the Binomial Series to find the first four terms of the Maclaurin series for $\sqrt[3]{1+x}$.

HINT Apply the Binomial Series with $r = \frac{1}{3}$, computing $\binom{1/3}{1}, \binom{1/3}{2}, \binom{1/3}{3}$ directly from the general coefficient formula.

CAUTION — before you solve Each exponent $r = \frac{1}{3}, \frac{1}{2}, -\frac{1}{2}, \dots$ gives its own distinct set of coefficients — don't reuse coefficients computed for a different value of r in an earlier problem.

Your working:

13. A series $\sum a_n$ has partial sums $S_n = \frac{2n}{n+1}$. Find $\sum_{n=1}^{\infty} a_n$ (i.e., find $\lim_{n \rightarrow \infty} S_n$), and state what this tells you about convergence.

HINT Recall that a series' sum is defined as $\lim_{n \rightarrow \infty} S_n$, so just take the limit of the given partial-sum formula directly.

CAUTION — before you solve There's no need to reconstruct the individual terms $a_n = S_n - S_{n-1}$ first — the question only asks for the sum, which is the partial-sum limit itself.

Your working:
